



National University of Computer and Emerging Sciences

Introduction to Course

AI-2002 Artificial Intelligence (SE)

Dr. Muhammad Bilal

Department of Software Engineering

Instructor Information

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Visiting Hours

Schedule and Office Hours (Spring-2026)

Days	8:30 AM 9:50 AM	10:00 AM 11:20 AM	11:30 AM 12:50 PM	1:00 PM 2:20 PM	2:20 PM 3:50 PM	3:55 PM 5:15 PM
Monday	Occupied					
Tuesday			Office Hour	AI (SE-A) C-410	AI (SE-B) C-408	
Wednesday				Office Hour	Office Hour	
Thursday			Office Hour	AI (SE-A) C-410	AI (SE-B) C-409	
Friday						

Text Book

- **Artificial Intelligence: A Modern Approach, 4th edition by S. Russell and P. Norvig, Prentice Hall, 2020. (Official slides provided by authors)**
 - Artificial Intelligence: Structures and Strategies for Complex Problem Solving, 6th ed. G. Luger, Addison Wesley, 2009
 - AI Algorithms, Data Structures, and Idioms in Prolog, Lisp and Java, G. Luger and W. Stubblefield, Addison Wesley, 2009
 - Artificial Intelligence: A Systems Approach. M. Tim Jones, Infinity Science Press, 2008
-

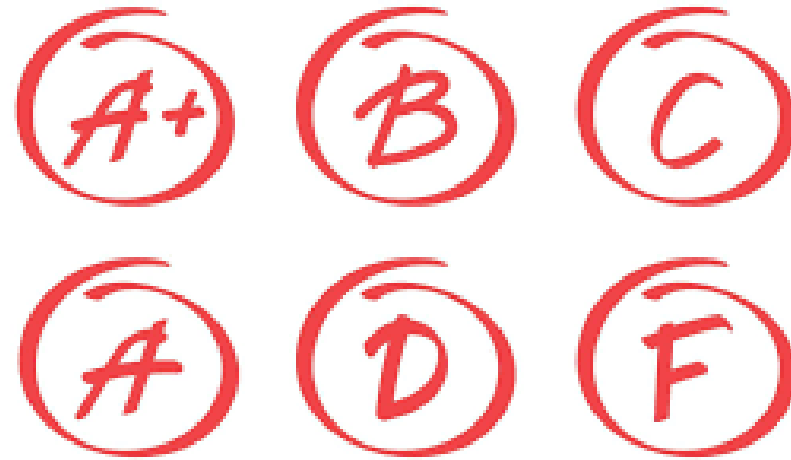
Tentative marks distribution

Assessment items	Weightage
• Quiz	12.5% (~4)
• Assignments	12.5% (~3)
• Project	10%
• Midterms(2)	30%
• Final-exam	35%

Late submissions will not be considered.
Avoid last minute submissions.

Grading policy

- Absolute grading



Course Plagiarism Policy

- Plagiarism in project or midterm/ final exam may result in F grade in the course.
 - Plagiarism in an assignment will result in zero marks in the whole assignments category.
-

Class Rules

- **Attendance:** Arrived **after 10 minutes** will be marked **late**.
- **Mobile Phone Usage:** Penalty (fine) as per university rules.
- **Eating:** Penalty (fine) as per university rules.

Course Materials

- Will be shared via **Google Classroom**

Class Code

ai7lcgat

Course Contents

List of Topics	Weeks
Introduction to AI	1
Intelligent Agents: Problem Solving, Agent types, Agent environments	1
Search problem formulation , Search space definition, Tree Search, Graph Search, Uninformed searches - BFS, DFS, UCS	1
Informed Search : Heuristic Function, Greedy Best First Search, and A* search. Problems with Search	1
Local Search & Optimization : Hill Climbing & Genetic Algorithm.	2
Game as Search Problems , Adversarial search Minimax and alpha beta pruning,	1

Course Contents

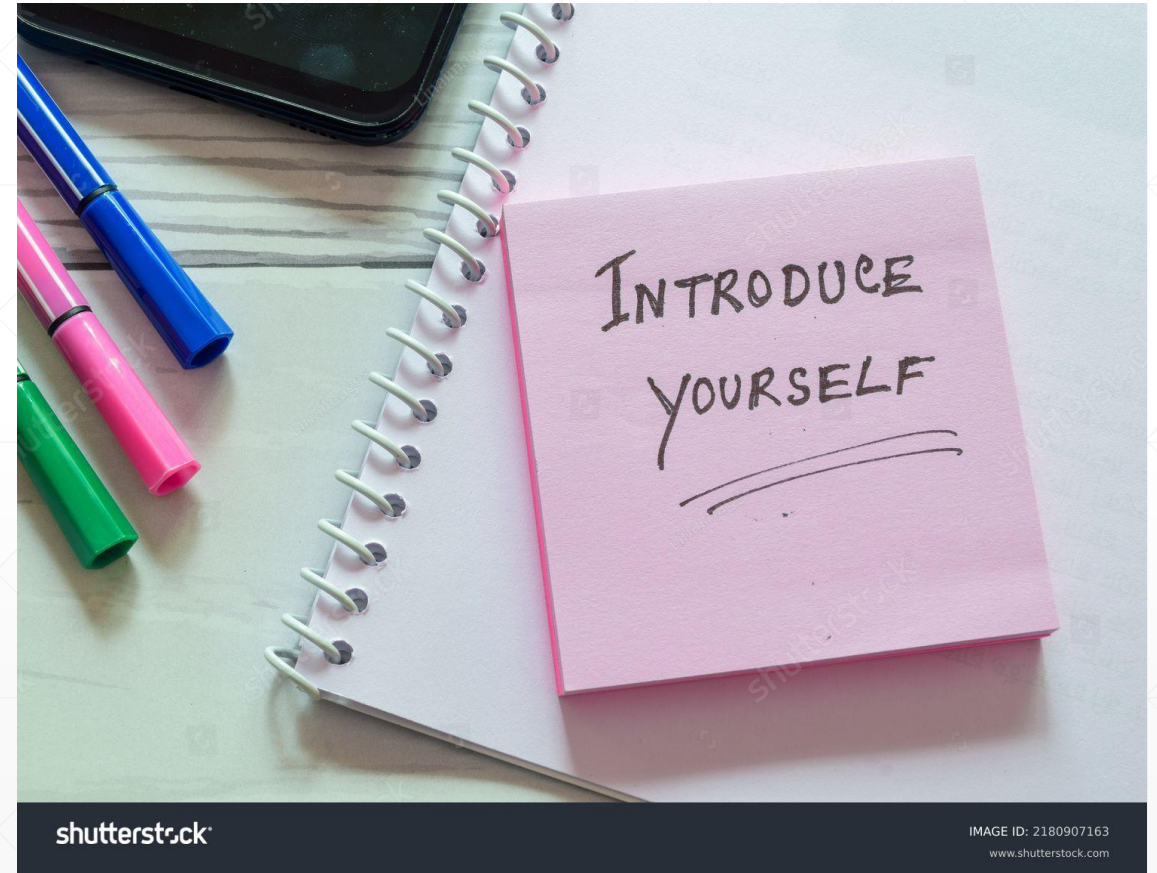
List of Topics	Weeks
Constraint Satisfaction Problems Back-tracking search for CSPs, Problem structure and problem decomposition, Local search for CSPs, Cryptarithmic Problem.	1
Intro to Machine Learning, Supervised Learning, KNN, Decision Tree	1
Unsupervised Learning, K-Means, Hierarchal Clustering, Naive Bayes	1
Neural Networks: Perceptron, Multilayer perceptron, Applications of neural networks, Perceptron Learning, Back-propagation, Gradient Descent	3
Intro to Reinforcement Learning, Q-Learning	1
Project Presentations	1

Announcements regarding AI (SE) course

- Only one show up for each exam would be arranged in a designated slot. All the students are encouraged to see their attempts in that show up.
- Missed quizzes and assignments will not be entertained.
- All the course material will be shared on google classroom.
- Google classroom is a formal forum for disbursement of educational material only.

It's Your Turn

- To Introduce Yourself
 - Your Name
 - Educational Background



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What is Artificial Intelligence?



We have heard about AI

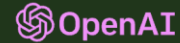
NEWS | 12 January 2023

Abstracts written by ChatGPT fool scientists

Researchers cannot always differentiate between AI-generated and original abstracts.



DALL-E 2 created this image in response to the text "teddy bears mixing sparkling chemicals as mad scientists in a steampunk style"



ChatGPT: Optimizing
Language Models
for Dialogue

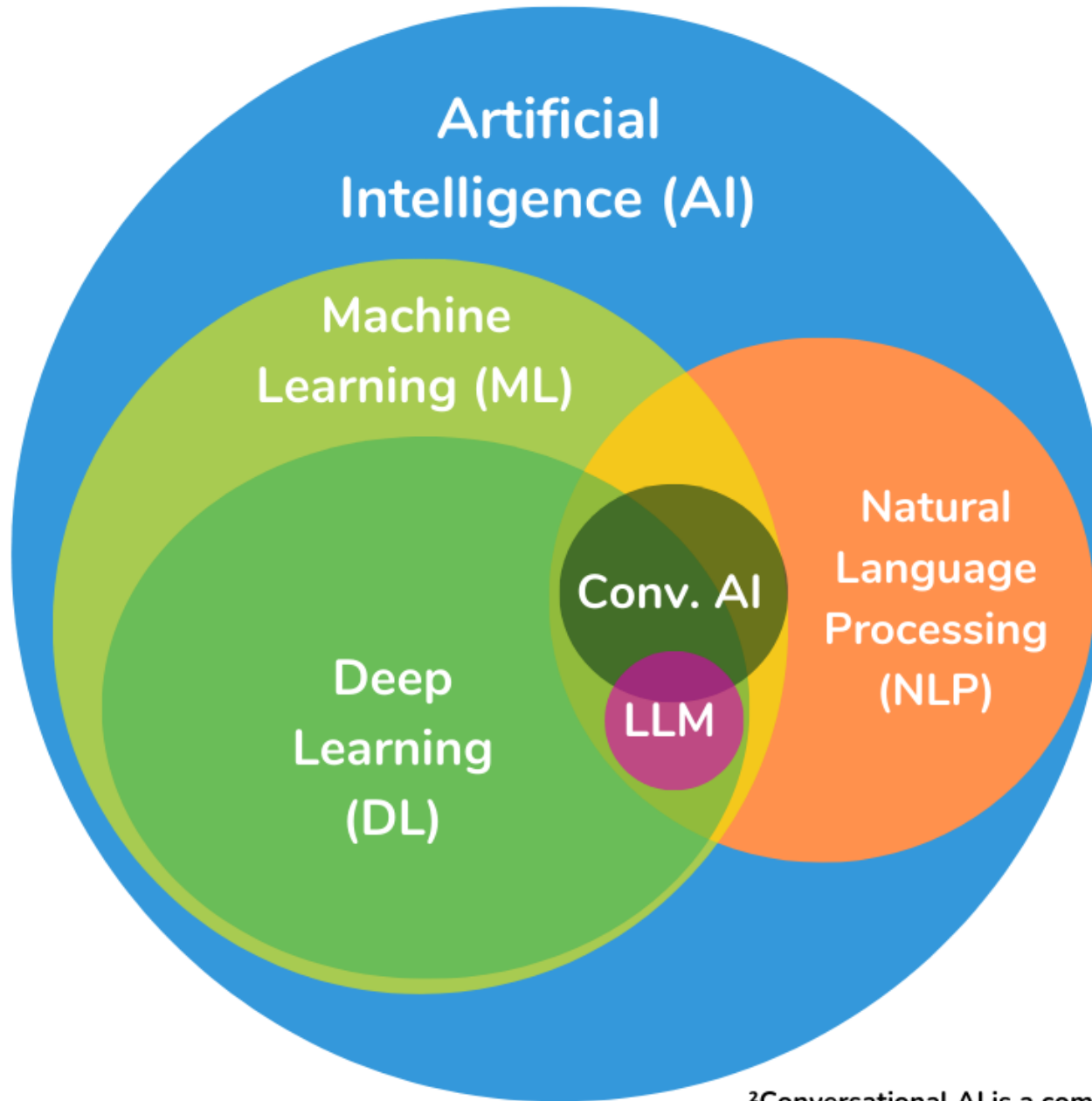


A broader (but vague) definition

- Artificial intelligence is the development and study of computer systems to address complex real-world problems typically associated with some form of intelligence.

What is AI/ML?

- Columbia's School of Engineering defines **Artificial Intelligence (AI)** as "the field of developing computers and robots that are capable of behaving in ways that both mimic and go beyond human capabilities".
- **Machine learning (ML)** is a subset of AI that uses algorithms to enable machines and computer systems to learn to recognize complex patterns and insights from data without explicit programming.
- The machine learns to recognize human speech, understand the meaning of texts, or even generate new content, for example, writing an essay.



- Artificial Intelligence (AI)
- Machine Learning (ML)
- Deep Learning (DL)
- Natural Language Processing (NLP)
- Large Language Model (LLM)¹
- Conversational AI (Conv. AI)²

¹LLM is an intersection of DL and NLP

²Conversational AI is a combination of ML and NLP. It may include DL and LLM, but that isn't always the case.

Can we have AI system without using Machine Learning?

- **Yes.** Simple AI systems can be explicitly programmed.
- Example: In the early 1960s, IBM developed and demonstrated "Shoebox", an innovative device capable of recognizing and responding to 16 spoken words, including the ten digits from "0" through "9".
- Rule-based expert system. These systems rely on predefined rules and logic to make decisions or solve problems, **rather than learning from data.**

ChatGPT

- OpenAI





Explain quantum computing in simple terms



Quantum computing is a type of computing where information is processed using quantum bits, or qubits, instead of classical bits. Qubits can exist in multiple states at the same time, unlike classical bits which can only be in one state (0 or 1) at a time. This allows quantum computers to perform certain types of calculations much faster than classical computers. However, they are still in the early stages of development and have not yet been able to perform tasks that would be useful for most practical applications.



AI in the “real world”

- AI is beyond controlled benchmarks and games

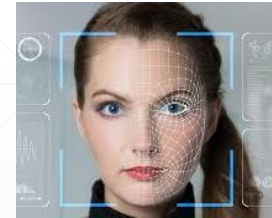
Translation



Spam detection



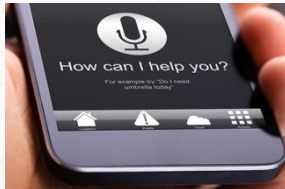
Facial recognition



Autonomous driving



Voice assistant



Stock market



Hiring systems



Natural Language

- Speech technologies (e.g. Siri)
 - Automatic speech recognition (ASR)
 - Text-to-speech synthesis (TTS)
 - Dialog systems
- Language processing technologies
 - Question answering
 - Machine translation
 - Web search
 - Text classification, spam filtering, etc...

"Il est impossible aux journalistes de rentrer dans les régions tibétaines"

Bruno Philip, correspondant du "Monde" en Chine, estime que les journalistes de l'AFP qui ont été expulsés de la province tibétaine du Qinghai "n'étaient pas dans l'illégalité".

Les faits Le dalaï-lama dénonce l'"enfer" imposé au Tibet depuis sa fuite, en 1959

Vidéo Anniversaire de la rébellion tibétaine : Le China sur ses gardes



"It is impossible for journalists to enter Tibetan areas"

Philip Bruno, correspondent for "World" in China, said that journalists of the AFP who have been deported from the Tibetan province of Qinghai "were not illegal."

Facts The Dalai Lama denounces the "hell" imposed since he fled Tibet in 1959

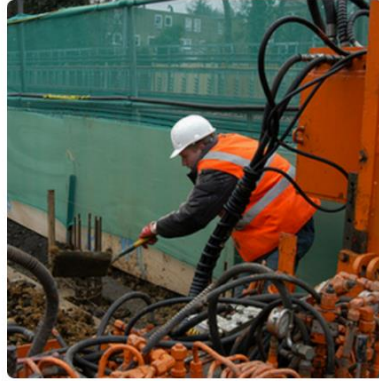
Video Anniversary of the Tibetan rebellion: China on guard



Computer Vision



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



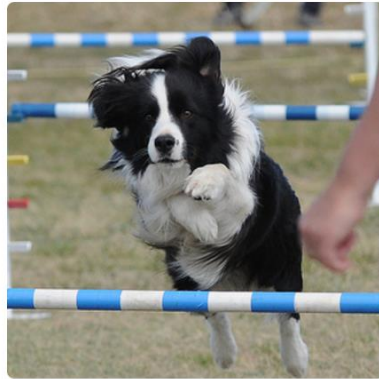
"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."

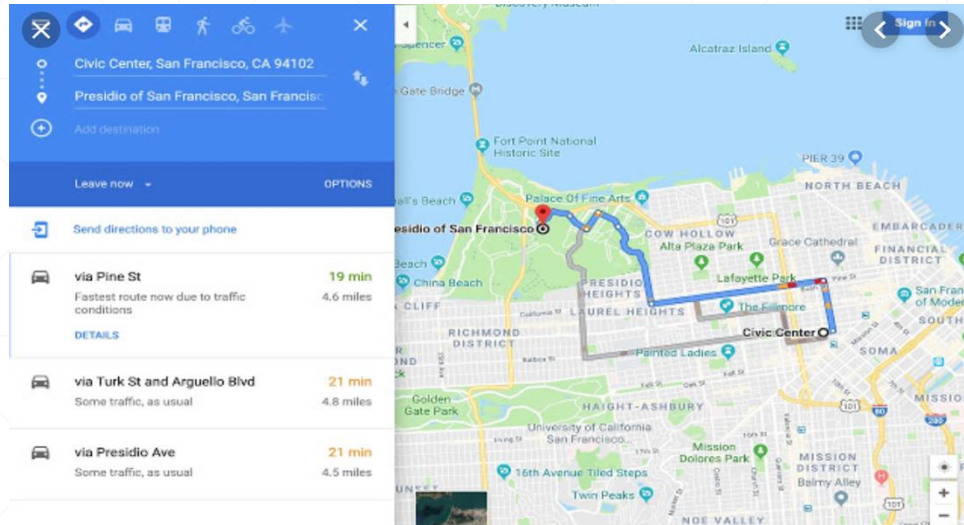


"young girl in pink shirt is swinging on swing."



"man in blue wetsuit is surfing on wave."

Tools for Predictions & Decisions

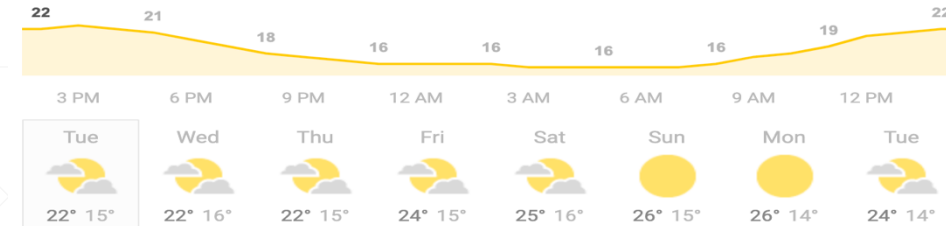


Berkeley, CA 94709
Tuesday 2:00 PM
Mostly Sunny

22°C | °F

Precipitation: 0%
Humidity: 58%
Wind: 24 km/h

Temperature Precipitation Wind



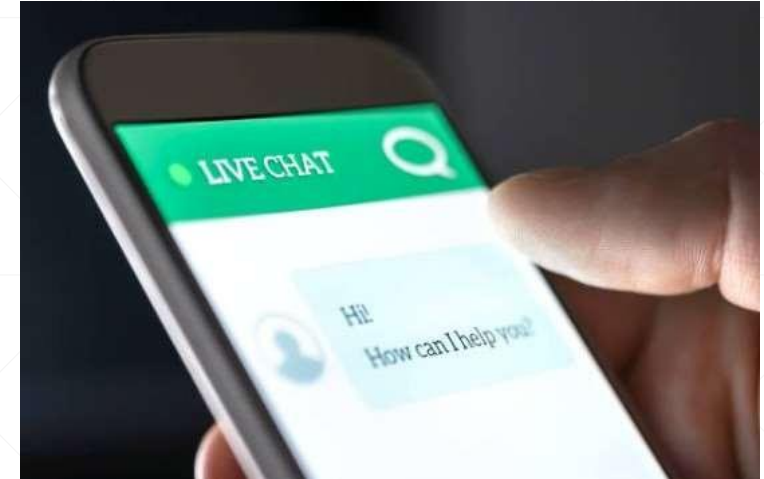
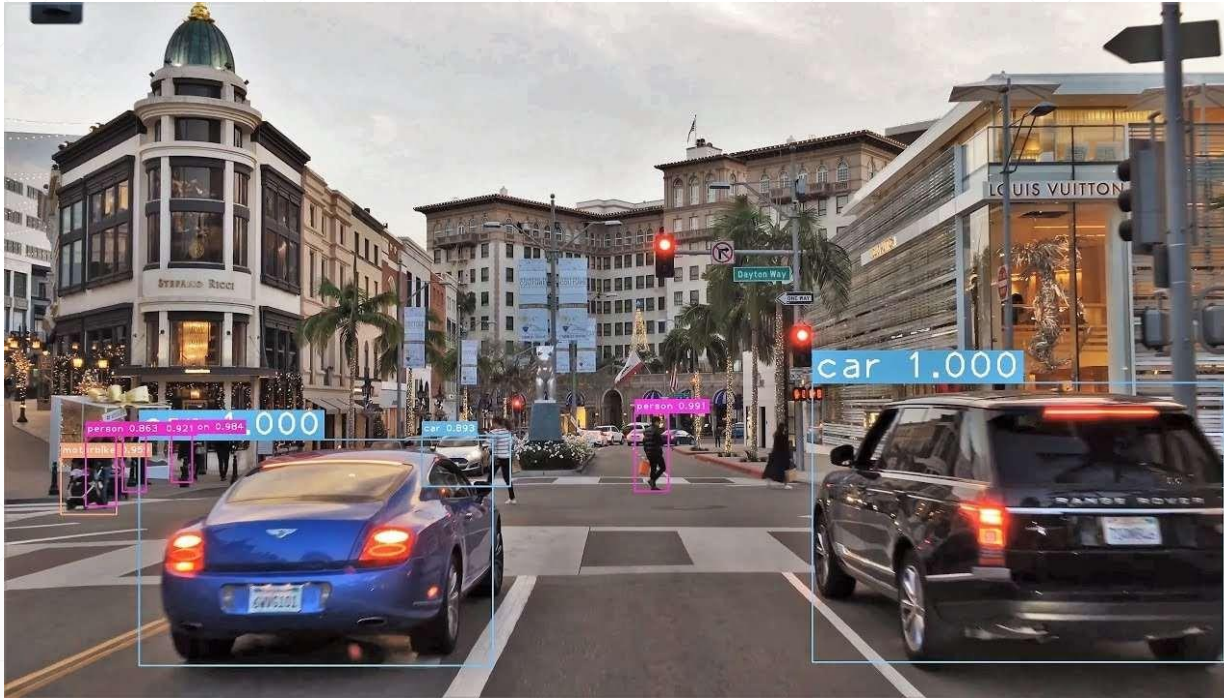
Robots



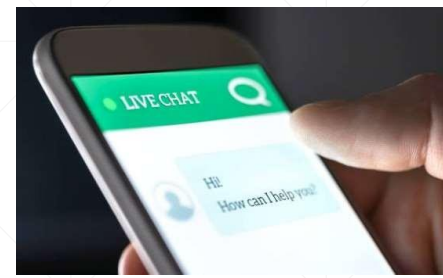
Human-AI Interaction



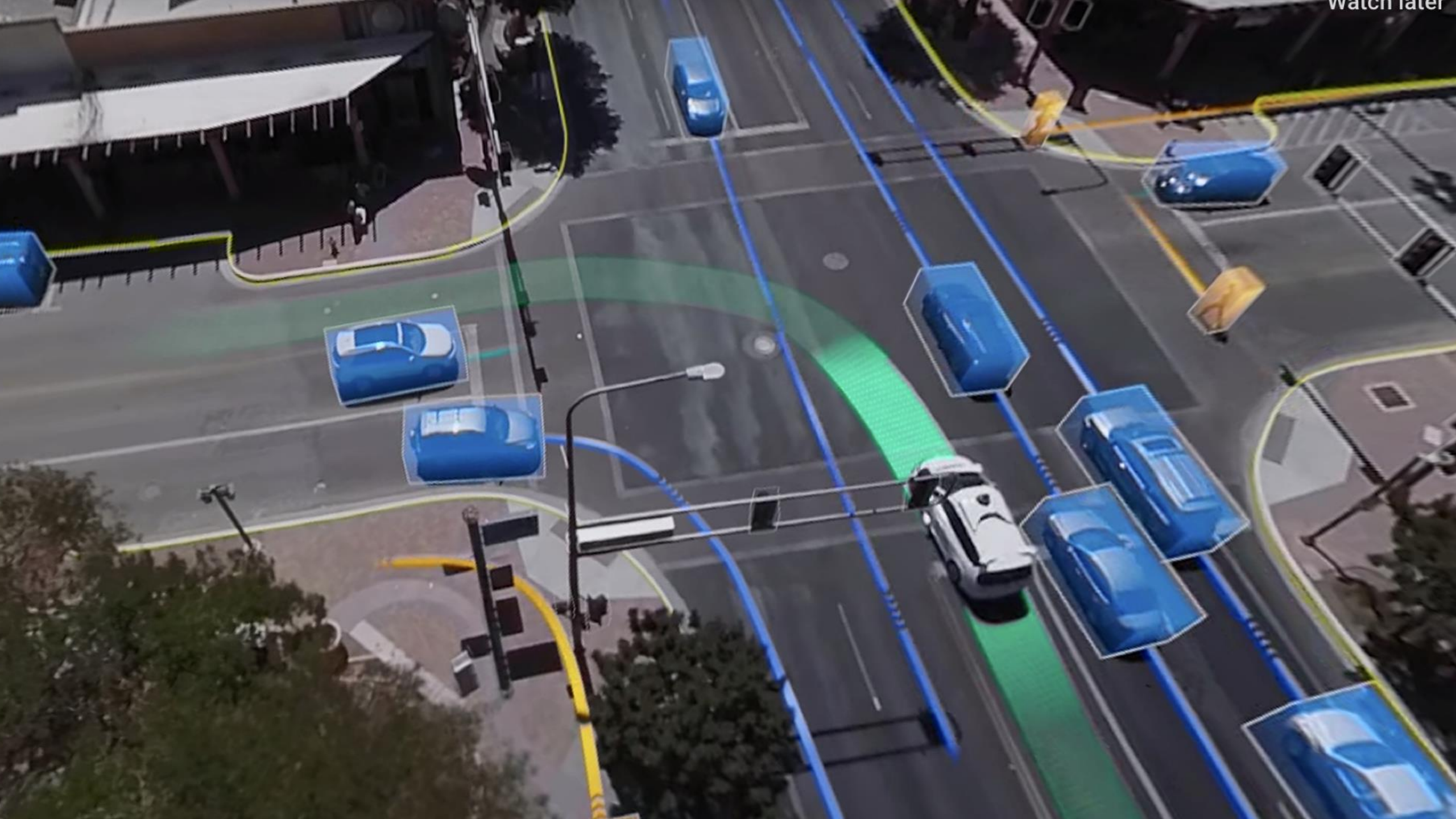
Complex real-world problems



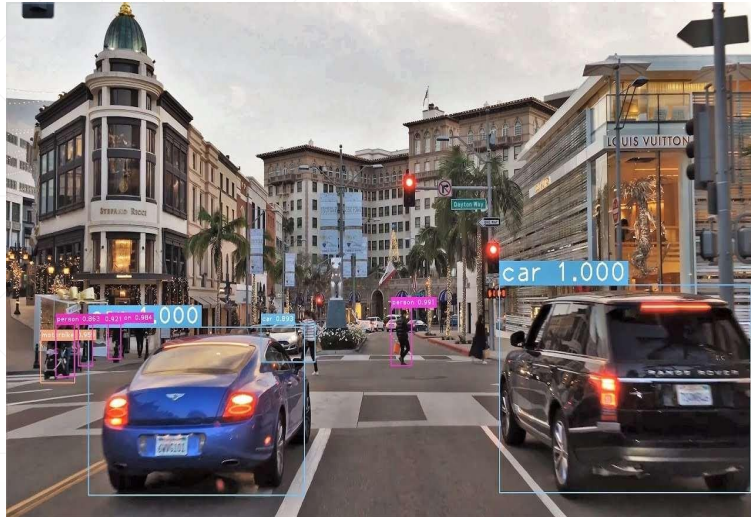
Strategy?!



- AI is about using technology to tackle complex real-world problems.
 - What are some complex real-world problems that come to mind?
 - Self-driving cars have to sense the complex scene in front of them (in real-time!) and make safe decisions that advance the car towards its destination.
 - Virtual assistants have to understand the rich subtleties of natural language, infer the user intent, and generate appropriate responses.
 - Medical systems have to interpret the subtle, noisy cues in medical images, take into account the messy incomplete patient information, and generate reliable diagnoses that can be trusted by doctors.
 - These are really really hard problems. How do you even begin to tackle them?
-



Bridging the gap

[illegible]

- At the end of the day, we need to write some code (and possibly build some hardware, too).
- But there is a huge gap between the problem and the solution.
- You don't want to just start jumping in and writing code without a clear plan of action.

Paradigm

Modeling

Inference

Learning

- We will adopt the **modeling-inference-learning** paradigm to help us navigate the solution space.
- There are three pillars, modeling, inference, and learning.
- In reality, the lines between the three pillars are blurry, but this paradigm serves as an ideal and a useful guiding principle.

Paradigm: Modeling

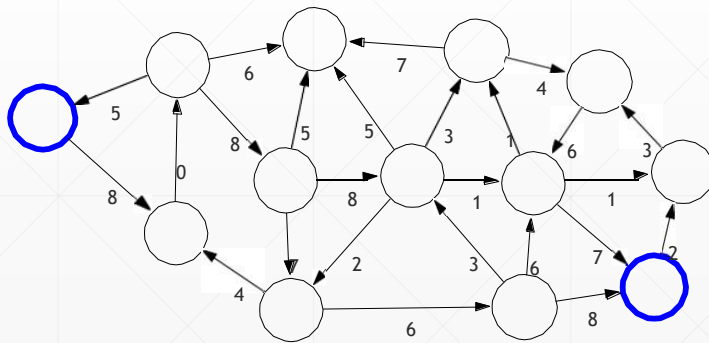
Real world



Modeling



Model

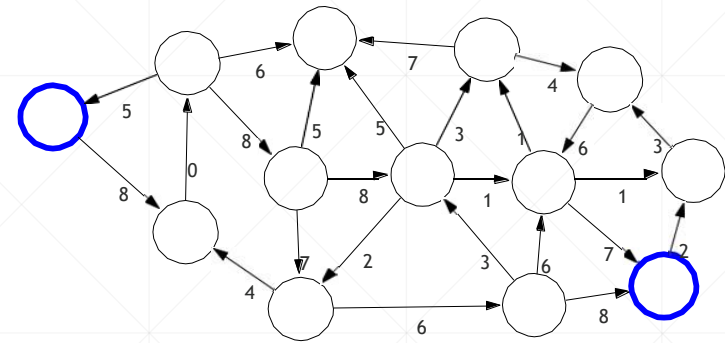


- The first pillar is modeling.
- Modeling takes messy real world problems and packages them into neat formal mathematical objects called **models**, which can be subject to rigorous analysis and can be operated on by computers.
- We might formulate the driving problem as a route finding problem as a graph where nodes represent points in the city, edges represent the roads, and the cost of an edge represents the traffic on that road.

Paradigm: Inference

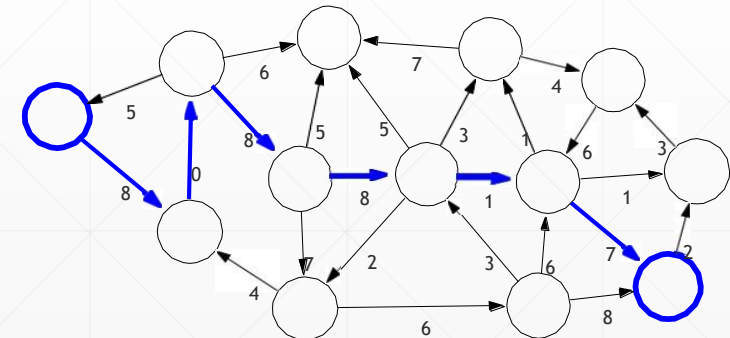
- The second pillar is inference. Given a model, the task of **inference** is to answer questions with respect to the model. For example, given the model of the city, one could ask questions such as: what is the shortest path? what is the cheapest path?
- The focus of inference is usually on efficient algorithms that can answer these questions.
- For some models, computational complexity can be a concern (games such as Go), and usually approximations are needed.

Model



Inference

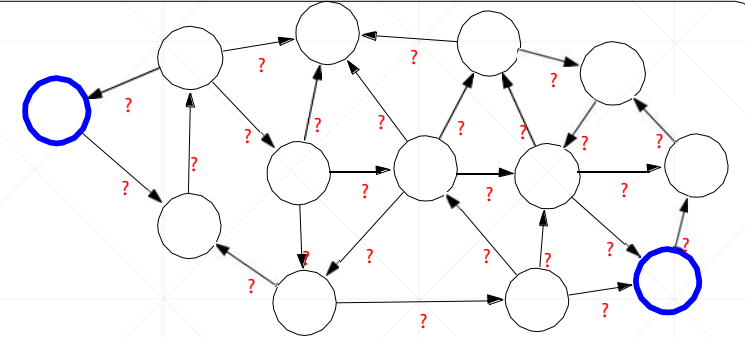
Predictions



Paradigm: Learning

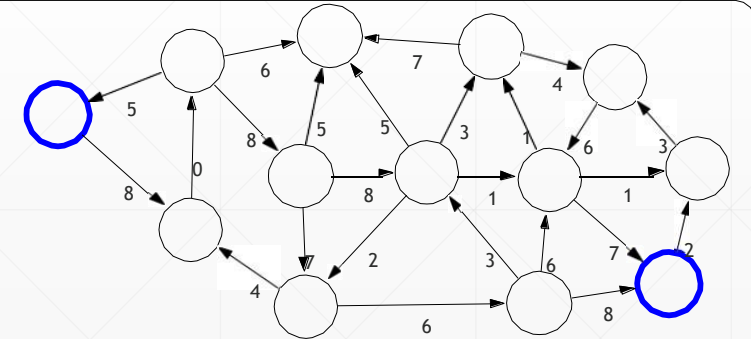
- The idea behind (machine) **learning** is to instead get it from data. Instead of constructing a model, one constructs a skeleton of a model (more precisely, a model family), which is a model without parameters. And then if we have the appropriate data, we can run a machine learning algorithm to tune the parameters of the model.
- The idea of learning is not tied to a particular model family (e.g., neural networks). Rather, it is more of a philosophy of how to produce models.

Model without
parameters



+data
Learning

Model with
parameters



What is Artificial Intelligence ?

		THOUGHT	
		HUMAN	RATIONAL
BEHAVIOUR	HUMAN	Systems that think like humans	Systems that think rationally
	RATIONAL	Systems that act like humans	Systems that act rationally

Some classic definitions

Think like a human

- Cognitive science / neuroscience
- Can't there be intelligence without humans?

Think rationally

- Logic and automated reasoning
- But, not all problems can be solved just by reasoning

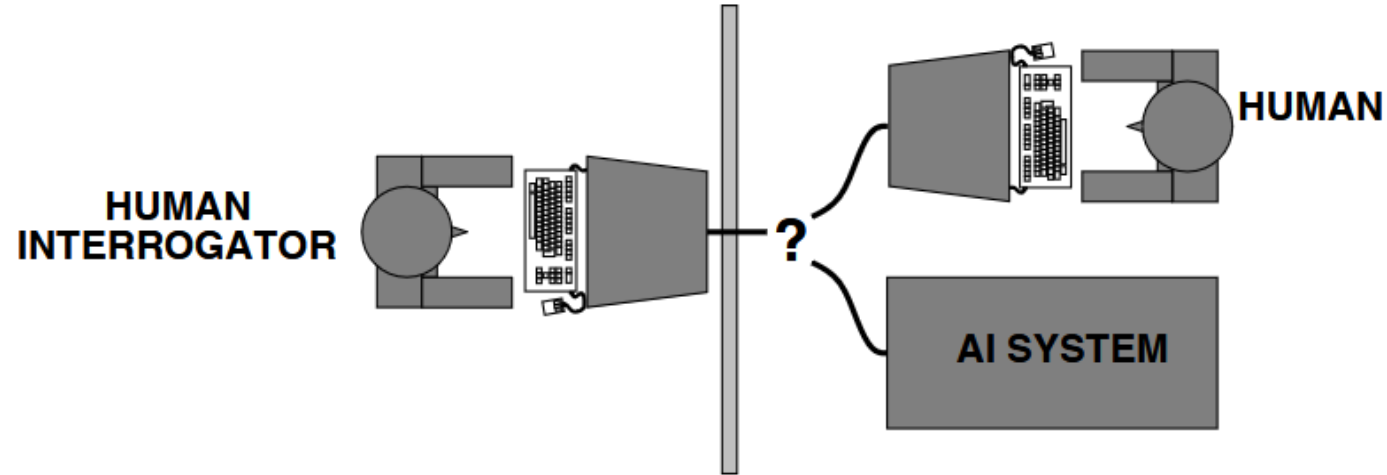
Act like a human

- Turing test
- ELISA, Loebner prize
- "What is 1228×5873 "? ... "I don't know, I'm just a human"

Act rationally

- Rational behavior: doing the right thing
- Basis for intelligence agents framework
- Unclear if this captures the current scope of AI research

Systems that act like humans



- You enter a room which has a computer terminal. You have a fixed period of time to type what you want into the terminal, and study the replies. At the other end of the line is either a human being or a computer system.
 - If it is a computer system, and at the end of the period you cannot reliably determine whether it is a system or a human, then the system is deemed to be intelligent.
-

- These cognitive tasks include:

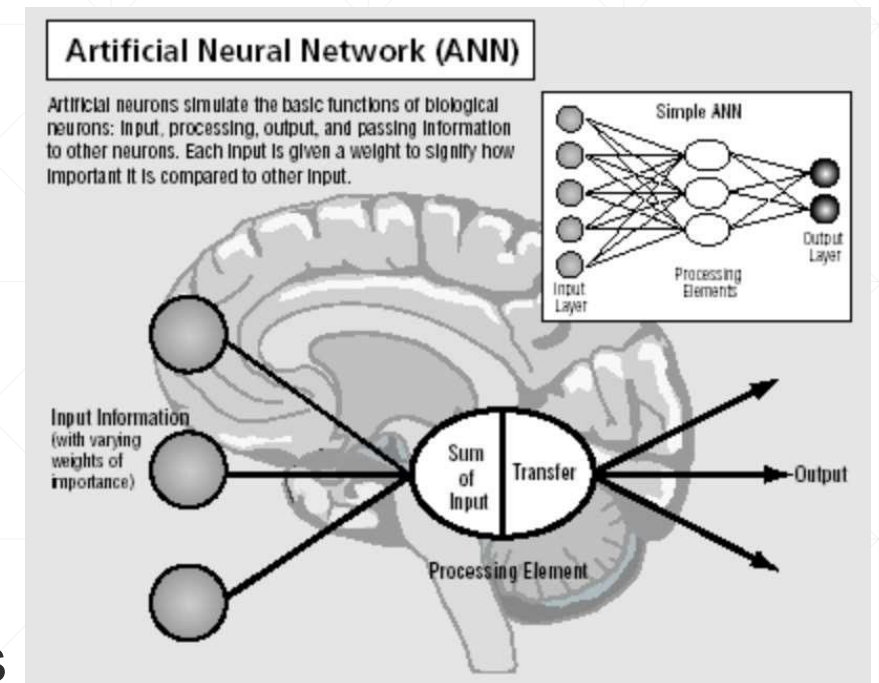
- Natural language processing
 - for communication with human
- Knowledge representation
 - to store information effectively & efficiently
- Automated reasoning
 - to retrieve & answer questions using the stored information
- Machine learning
 - to adapt to new circumstances

- Includes two more issues:

- *Computer vision*
 - to perceive objects (seeing)
 - *Robotics*
 - to move objects (acting)
-

Systems that think like humans: Cognitive Modeling

- Humans as observed from ‘inside’
- How do we know how humans think?
 - Introspection vs. psychological experiments
- Cognitive Science
- “The exciting new effort to make computers think ... machines with minds in the full and literal sense” (Haugeland)
- “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning...” (Bellman)



Acting rationally and Rational agents

- Rational behavior: doing the right thing
- The right thing: that which is expected to maximize goal achievement, given the available information
- Doesn't necessarily involve thinking e.g., blinking reflex but thinking should be in the service of rational action
- An agent is an entity that perceives and acts.
- Artificial Intelligence is about designing rational agents.
- Abstractly, an agent is a function from percept histories to actions. $f : P^* \rightarrow A$

Questions